

1. INTEGRATED SITUATION:

TZW LTD is a company dedicated to managing, inspecting, and maintaining fire safety equipment, particularly fire extinguishers, across various large commercial and industrial facilities in the region. Currently, their Fire Extinguisher Management System is built using a Monolithic Architecture but is facing several challenges, including missed inspection deadlines, difficulties in tracking maintenance history, and compliance issues. To enhance efficiency and functionality, TZW LTD seeks to upgrade their system to Microservices Architecture. This new system will allow users to check extinguisher statuses, schedule inspections, log maintenance actions, track compliance, and generate real-time reports.

Task:

As a Full Stack Developer, you have been hired to analyze, design, and implement a Fire Extinguisher Management System based on RESTful microservices, ensuring scalability, maintainability, and high availability.

Activities:

Activity 1: Requirement Analysis and Design

1. Identify and define the required microservices and RESTful API contract using OpenAPI/Swagger documentation
2. Define and design database model
3. Design mockup for user registration/signup form using Figma or other tools

Activity 2: User management services

a) Define user roles (Admin, Inspector and User)

- Admin: Manages overall system features, user accounts, and data integrity.
- Inspector: Responsible for conducting inspections, logging results, and scheduling maintenance.
- User: Can view extinguisher status and schedule inspections. Create user experience

b) Implement User Registration Endpoint:

- Create an API for new users to register, receiving and validating the following:
 - First Name
 - Last Name
 - Email
 - Password

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- c) Implement Authentication Endpoint and apply JWT-based authentication and role-based authorization: Develop an API for user login and logout, managing sessions or tokens securely.
- d) Develop user profile management (update their profile, change their passwords, and recover their passwords).

Activity 3: Fire Extinguishers Management Services

a) Create endpoints to Register new extinguisher with below records:

- Serial Number
 - Location
 - Type (Water, CO2, Foam, Dry Chemical)
 - Size (2.5 lbs., 5 lbs., 9 lbs., 12 lbs.,)
 - Installation Date
 - Expiry date
 - Status
- b) Create endpoints to List all extinguishers
 - c) Create endpoints to View extinguisher details by id
 - d) Create endpoints to Update extinguisher information
 - e) Create endpoints to Remove extinguisher record

f) Develop an API to allow users to schedule inspections, which includes:

- Selecting the extinguisher
- Choosing a date and time
- Notifying relevant personnel

g) Implement an API for inspectors to log maintenance activities, including:

- Actions taken
- Date of the action
- Conditions noted during the maintenance

Activity 4: Reporting Services

1 - Generate real-time reports:

- Total number of extinguisher in the stock (daily, monthly, yearly)
- Inspection status